EDUSAMBAM ASSESSMENT SERIES

Biology Comprehensive Practice Test 5

PRINTABLE PRACTICE WORKSHEET

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Candidate Name:		**Roll No.:**	
Class / Grade:		**Date:**	
School / Institution:		**Test Centre:**	

Subject	Biology	**Topic**	Biology Comprehensive Practice Test 5	**Total Marks**	100
Total Questions	45	**Time Allowed**	1 Hour (60 minutes)	**Version**	A

GENERAL INSTRUCTIONS

1. Read each question carefully before answering.
2. For Section A and Section B, mark one answer only.
3. For Sections C and E, write your answer clearly in the space provided.
4. For Section D, write the correct letter from Column B against each item in Column A.
5. Do not use correction fluid. Keep all answers clear and legible.
6. This worksheet is prepared from the EDUSAMBAM practice assessment supplied for this paper.

PAPER PATTERN

Section	Questions	Marks Each	Total
Section A — Multiple Choice	15	2	30
Section B — True / False	10	2	20
Section C — Fill in the Blank	10	2	20
Section D — Column Matching	5 pairs	2	10
Section E — Short Answer	5	4	20
Total	**45**	**—**	**100**

Section A — Multiple Choice — 30 marks

1. Which statement about asexual reproduction is correct? — Basic · 2 pts

A. It involves one parent and produces offspring genetically identical to the parent (apart from mutations)
B. It requires the fertilisation of an egg by a sperm
C. It always produces offspring with new gene combinations
D. It occurs only in animals

Answer: ☐A ☐B ☐C ☐D

2. Fertilisation is: — Basic · 2 pts

A. Division of a zygote into two cells
B. Release of an egg from the ovary
C. Growth of a seed into a seedling
D. The fusion of a male gamete and a female gamete to form a zygote

Answer: ☐A ☐B ☐C ☐D

3. Which type of cell division produces new cells for growth and repair? — Basic · 2 pts

A. Meiosis
B. Mitosis
C. Fertilisation
D. Pollination

Answer: ☐A ☐B ☐C ☐D

4. In a flowering plant, pollination is the transfer of pollen from an anther to a: — Basic · 2 pts

A. Root hair
B. Seed coat
C. Stigma
D. Leaf stoma

Answer: ☐A ☐B ☐C ☐D

5. In a human pregnancy, the placenta: **Basic · 2 pts**

A. Produces eggs
B. Stores sperm
C. Digests food for the baby
D. Allows exchange of oxygen, nutrients and wastes between maternal and fetal blood

Answer: ☐A ☐B ☐C ☐D

6. Which set of conditions does a seed need in order to germinate? **Intermediate · 2 pts**

A. Light, soil and carbon dioxide only
B. Water, oxygen and a suitable temperature
C. Sunlight and minerals only
D. Pollinators and nectar

Answer: ☐A ☐B ☐C ☐D

7. Cell differentiation is the process in which: **Intermediate · 2 pts**

A. Cells become specialised for particular functions
B. Cells join together to form a zygote
C. Cells release gametes
D. Cells divide without copying DNA

Answer: ☐A ☐B ☐C ☐D

8. A zygote is: **Intermediate · 2 pts**

A. A haploid sperm cell
B. A type of seed
C. The diploid cell formed by fertilisation
D. The structure that nourishes the fetus

Answer: ☐A ☐B ☐C ☐D

9. The main site of photosynthesis in plant cells is the: **Intermediate · 2 pts**

A. Nucleus
B. Chloroplast
C. Lysosome
D. Ribosome

Answer: ☐A ☐B ☐C ☐D

10. Photosynthesis converts light energy into: **Intermediate · 2 pts**

A. Chemical energy stored in organic molecules
B. Heat energy that is lost
C. Sound energy
D. Kinetic energy only

Answer: ☐A ☐B ☐C ☐D

11. Carbon dioxide enters most leaves mainly through: **Advanced · 2 pts**

A. Xylem vessels
B. Root hairs
C. Phloem sieve tubes
D. Stomata

Answer: ☐A ☐B ☐C ☐D

12. Xylem primarily transports: **Advanced · 2 pts**

A. Sugars made in the leaves
B. DNA
C. Water and mineral ions
D. Oxygen to mitochondria

Answer: ☐A ☐B ☐C ☐D

13. Phloem transports: **Advanced · 2 pts**

A. Organic substances such as sucrose from sources to sinks
B. Only water
C. Only mineral ions
D. Only carbon dioxide

Answer: ☐A ☐B ☐C ☐D

14. If a plant's stomata close, a likely immediate effect is: **Advanced · 2 pts**

A. Increased carbon dioxide entry
B. Elimination of all xylem flow
C. A guaranteed increase in photosynthesis
D. Reduced carbon dioxide entry and reduced water loss

Answer: ☐A ☐B ☐C ☐D

15. A plant with adequate light but very low magnesium may have reduced photosynthesis because: **Advanced · 2 pts**

A. Magnesium replaces carbon dioxide
B. Magnesium is needed to make chlorophyll
C. Magnesium is the main sugar in the leaf
D. Magnesium closes every stoma permanently

Answer: ☐A ☐B ☐C ☐D

Section B — True / False — 20 marks

16. Sexual reproduction usually produces offspring that are genetically different from their parents. **Basic · 2 pts**

A. True
B. False

Answer: ☐A ☐B

17. Mitosis produces genetically different gametes. **Basic · 2 pts**

A. True
B. False

Answer: ☐A ☐B

18. Pollination and fertilisation are the same process in plants. **Basic · 2 pts**

A. True
B. False

Answer: ☐A ☐B

19. A human fetus develops inside the mother's uterus. Basic · 2 pts
A. True **B.** False
Answer: ☐ A ☐ B

20. Growth in many multicellular organisms involves both cell division and cell enlargement. Intermediate · 2 pts
A. True **B.** False
Answer: ☐ A ☐ B

21. Plants require carbon dioxide for photosynthesis. Intermediate · 2 pts
A. True **B.** False
Answer: ☐ A ☐ B

22. Photosynthesis occurs only in roots. Intermediate · 2 pts
A. True **B.** False
Answer: ☐ A ☐ B

23. Xylem is associated mainly with water and mineral transport. Advanced · 2 pts
A. True **B.** False
Answer: ☐ A ☐ B

24. Transpiration is identical to photosynthesis. Advanced · 2 pts
A. True **B.** False
Answer: ☐ A ☐ B

25. Increasing light intensity always increases the rate of photosynthesis without limit. Advanced · 2 pts
A. True **B.** False
Answer: ☐ A ☐ B

Section C — Fill in the Blank 20 marks

26. Sperm are produced in the male organs called the _____. Intermediate · 2 pts
Answer: _____

27. In females, eggs are released from the _____. Intermediate · 2 pts
Answer: _____

28. In humans the organ in which the fetus develops is the _____. Intermediate · 2 pts
Answer: _____

29. Unspecialised cells that can develop into many different cell types are called _____ cells. Intermediate · 2 pts
Answer: _____

30. The male gametes of flowering plants are found inside grains called _____. Intermediate · 2 pts
Answer: _____

31. The green pigment that absorbs light for photosynthesis is _____. Advanced · 2 pts
Answer: _____

32. Stomata are opened and closed by a pair of _____ cells. Advanced · 2 pts
Answer: _____

33. Loss of water vapour from leaves is called _____. Advanced · 2 pts
Answer: _____

34. Movement of sugars through phloem from sources to sinks is called _____. Advanced · 2 pts
Answer: _____

35. The tissue beneath the upper epidermis that contains many chloroplasts is the _____ mesophyll. Advanced · 2 pts
Answer: _____

Section D — Column Matching

10 marks

36–40. Match each term in Column A with the correct description in Column B. **Advanced · 10 pts**

Column A	Answer	Column B
Gamete	____	**A.** Cell division that makes identical cells for growth and repair
Mitosis	____	**B.** Transports sugars from sources to sinks
Pollination	____	**C.** Transports water and mineral ions
Xylem	____	**D.** Haploid sex cell used in reproduction
Phloem	____	**E.** Transfer of pollen to a stigma

Section E — Short Answer

20 marks

41. Compare sexual and asexual reproduction. **Advanced · 4 pts**

42. Describe the main stages from fertilisation to the development of a human fetus. **Advanced · 4 pts**

43. Explain why seeds need water, oxygen and a suitable temperature to germinate. **Advanced · 4 pts**

44. Explain how stomata create a trade-off between carbon dioxide uptake and water conservation. **Advanced · 4 pts**

45. Explain why xylem and phloem are functionally different transport systems. **Advanced · 4 pts**

Candidate Signature: _____ **Invigilator Signature:** _____

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Marks Obtained: / 100	Checked By:
Result / Grade:	Date Checked:

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Biology Comprehensive Practice Test 5 — Answer Key

Teacher reference: equivalent grammatically correct wording is acceptable for open-ended items when the meaning is preserved.

Section A — Multiple Choice

1. A
2. D
3. B
4. C
5. D
6. B
7. A
8. C
9. B
10. A
11. D
12. C
13. A
14. D
15. B

Section B — True / False

16. True
17. False
18. False
19. True
20. True
21. True
22. False
23. True
24. False
25. False

Section C — Fill in the Blank (accepted answers)

26. testes / testis
27. ovary / ovaries
28. uterus / womb
29. stem
30. pollen
31. chlorophyll
32. guard
33. transpiration
34. translocation
35. palisade

Section D — Column Matching

36. Gamete — D
37. Mitosis — A
38. Pollination — E
39. Xylem — C
40. Phloem — B

Section E — Short Answer (Model Answers)

41. Asexual reproduction needs one parent and produces offspring that are genetically identical clones, and it can be fast because no mate is needed. Sexual reproduction uses two parents, whose gametes (made by meiosis) fuse at fertilisation, producing genetically varied offspring, which helps a population cope with a changing environment.

Marking: 1 mark: asexual needs one parent; sexual uses two parents or two gametes; 1 mark: asexual offspring are identical clones; sexual offspring vary; 1 mark: sexual reproduction involves meiosis and fertilisation; 1 mark: a stated advantage (asexual is fast; variation helps survival in change).

42. A sperm fertilises an egg in the oviduct to form a zygote. The zygote divides repeatedly by mitosis to form a ball of cells (an embryo), which implants in the lining of the uterus, where a placenta forms. The cells then differentiate into specialised tissues and organs and the fetus grows.

Marking: 1 mark: a sperm fertilises an egg to form a zygote; 1 mark: the zygote divides by mitosis into a ball of cells (embryo); 1 mark: the embryo implants in the uterus lining and a placenta forms; 1 mark: cells differentiate into tissues and organs as the fetus grows.

43. Water is taken in so the seed swells and its enzymes become active and begin to break down the stored food. Oxygen is needed for aerobic respiration, which releases energy for growth. A suitable warm temperature lets the enzymes work well. The stored food then supports growth of the root and shoot until the seedling can photosynthesise.

Marking: 1 mark: water is absorbed, making the seed swell and activating enzymes; 1 mark: oxygen is needed for aerobic respiration to release energy; 1 mark: a warm temperature lets enzymes work well; 1 mark: stored food supports growth of the root and shoot.

44. Open stomata let carbon dioxide diffuse into the leaf for photosynthesis, but they also let water vapour escape by transpiration. Closing them saves water but reduces carbon dioxide entry and so slows photosynthesis. Guard cells adjust the opening to balance these needs, for example closing stomata in hot, dry conditions.

Marking: 1 mark: open stomata let carbon dioxide in for photosynthesis; 1 mark: open stomata also let water vapour escape (transpiration); 1 mark: closing them saves water but reduces carbon dioxide entry and photosynthesis; 1 mark: guard cells regulate the opening to balance the needs.

45. Xylem carries water and mineral ions from the roots up to the leaves, mainly in one direction, pulled by transpiration through dead, hollow, lignified vessels. Phloem carries sugars such as sucrose from sources, like leaves, to sinks, like roots, fruits or growing tips, through living sieve tubes supported by companion cells, and it can move materials in different directions.

Marking: 1 mark: xylem carries water and mineral ions; 1 mark: phloem carries sugars such as sucrose; 1 mark: xylem moves materials upward; phloem moves them from sources to sinks; 1 mark: they differ in structure or driving force (dead hollow vessels and transpiration pull; living sieve tubes).